Life Science Use Case

A Practical Hands-On Workshop using SAS for a Clinical Trial Analysis

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Clinical Trial Analysis

This use case is designed for beginners to learn the basic and essential steps for analyzing Clinical Trials with the enterprise SAS software in a highly regulated environment. No prior knowledge, e.g. SAS, Programming, is necessary. All analysis are created using a no-code, low-code approach.

The Use Case includes different parts along the typically steps of a Data Science Project:

- Access / Loading Data
- Investigate / Clean / Transform Data
- Reporting / Dashboards
- Descriptive Analysis
- Creating Analytical Models
- Survival Analysis

With the skills learned in this hands-on workshop, participants will then be able to perform analysis independently in SAS Visual Analytics. Students use the free of charge SAS Skill Builder e-learning and the included SAS Viya for Learners environment. Working through the use case takes approximately 90 - 120 minutes.

Clinical Trial Analysis

Business Context

You are a biostatistician and you work for an important life science company, which is about to launch a new product SASULIN on the market.

On that regard, the company need to conduct a clinical trial to ensure our products safety and approval by the regulating authorities.

In your role as a biostatistician, with no programming skills, your task is to analyze the given data from the clinical trial and answer important questions about the safety of the new product SASULIN.

Data Dictionary

Four data sets, contains several information and observations about the patients who took part in the clinical trial. The data are available in ADAM and CDISC SDTM format, a data format established in the clinical environment.

Table: ADTTE
Description:
26 Variables, 254 Rows

Excerpt

Variable	Description
AVAL	is the elapsed time to the event of interest from the origin. For example, if AVAL is measured in days, AVAL would be ADT – STARTDT or ADT – STARTDT + 1
CNSR	is a required variable for a time-to-event analysis dataset, though it is a conditionally required variable with respect to the ADaM BDS. For example, CNSR = 0 for event and CNSR > 0 for censored records.
TRTP	is a record-level identifier that represents the planned treatment attributed to a record for analysis purposes.

Data Dictionary

Table: DM_DM

Description: Demographic Data

28 Variables, 60 Rows

Excerpt

Variable	Description
USUBJID	Patient ID
ARM / ARMCD	planed treatment the subject taken (ARMCD if you want to use the code instead of the label)
ACTARM/ACTARMCD	actual treatment the subject taken (ACTARMCD if you want to use the code instead of the label)
RACE	Race
SEX	Gender
SITEID	Site Number

Data Dictionary

Table: AE_DM

Description: Adverse Events

54 Variables, 40 Rows

Excerpt

Variable	Description
RFSTDTC / RFENDTC	indicate a patient's initial and final exposure to a study drug or treatment, crucial for analyzing patient involvement and events within the study. Understanding the patient's study involvement timeline is crucial for event analysis.
RFXSTDTC / RFXENDTC	variables specifically represent the date and time of the first and last study exposure to the protocol-specified treatment or therapy, respectively.
RFICDTC	The earliest date of informed consent among any subject (Date/Time of Informed Consent) that enrolled in the study.
RFPENDTC	Final date on which data was collected for any subject (Date/Time of End of Participation, RFPENDTC) based on the protocol.
PATIENTID	Patient ID
AGE	Age
SEX	Gender
AESEV	Severity
AEBODSYS	The body system (also know as system organ class) that is affected.
AEHLGT	coded using the Medical Dictionary for Regulatory Affairs (MedDRA). Sponsors should code each term to the MedDRA-specified hierarchy of terms
AEDECOD	dictionary-derived text description; equivalent to the Preferred Term (PT in MedDRA). The sponsor is expected to provide the dictionary name and version used to map the terms utilizing the define.xml external codelist attributes.
AEREL	Adverse Event Relativity

Data Dictionary

Table: DIABETES

Description: Linked patient data between treatment and Diabetes

9 Variables, 768 Rows

Variable	Description
OUTCOME	Binary, does the drug have an impact on the patient.
AGE	Age
BloodPressure	Blood Pressure
BMI	Body Mass Index
DiabetesPedigreeFunction	Diabetes Pedigree
Glucose	Glucose
Insulin	Insulin
Pregnancies	Pregancy
SkinThickness	Skin thickness

Clinical Trial Analysis

Goal / Questions

Reporting of the Patient Demography Distribution

- How many patients are engaged / enrolled in that clinical trial ?
- What is the number of patients by type of drugs?
- What is the split of patient by age category?
- How is gender impacting the number of patients?
- What are the impacts of race or site number?

Adverse Events this Clinical Trial is potentially causing

- Which patients facing an Adverse Event?
- How does the event are across?
- What is the investigator's opinion as to the causality of the event to the treatment?
- What Adverse Event's are qualified as serious or not?

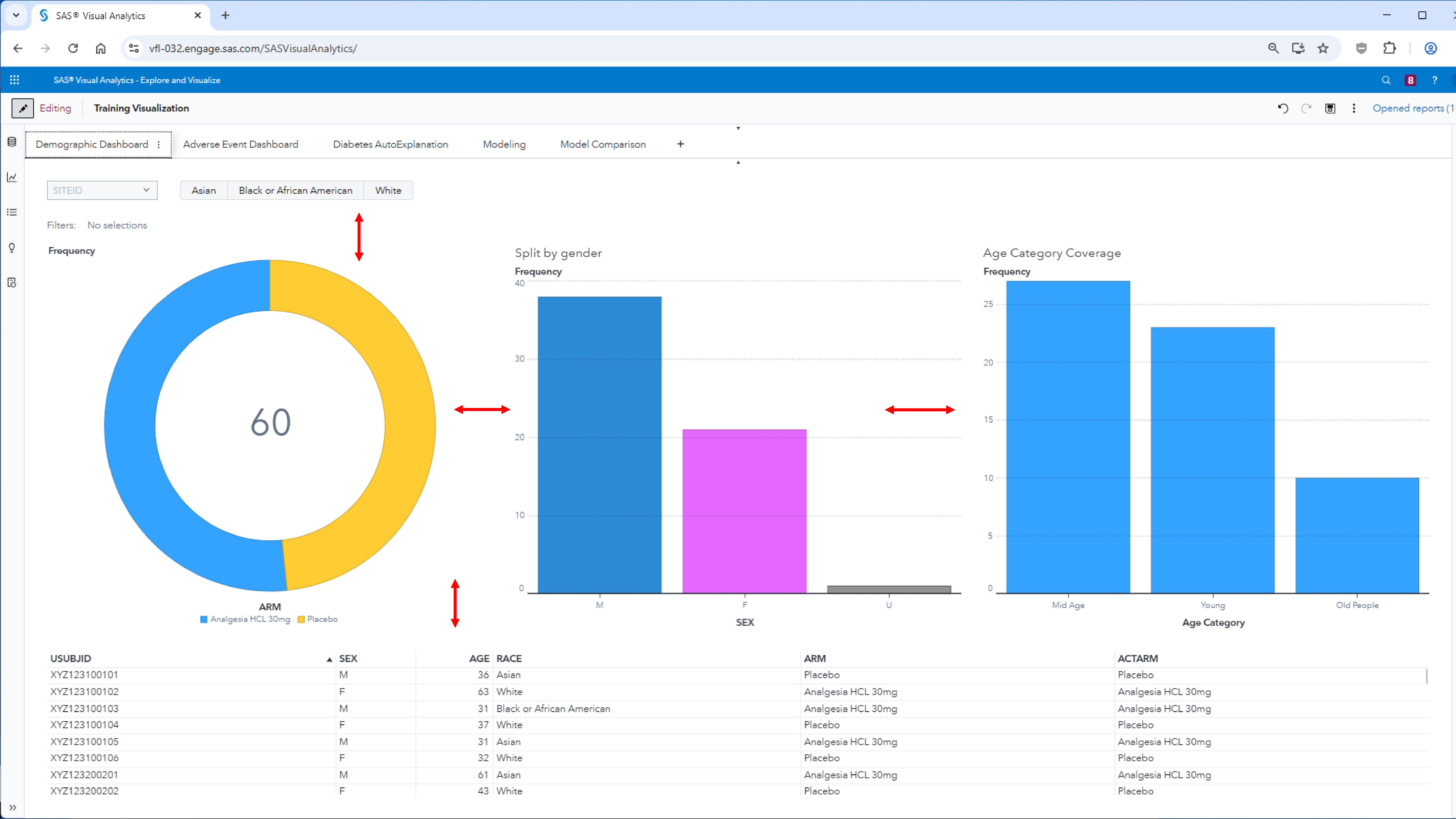
Prediction of impact of patients

- What are the main factors of impact?

Survial Analysis

- What is the proportion of a population which will survive past a certain time?
- Of those that survive the main event, at what rate will they die?
- Can multiple causes of death be taken into account?
- How do particular circumstances or characteristics increase or decrease the probability of survival?

Demographic Dashboard


SAS® Visual Analytics
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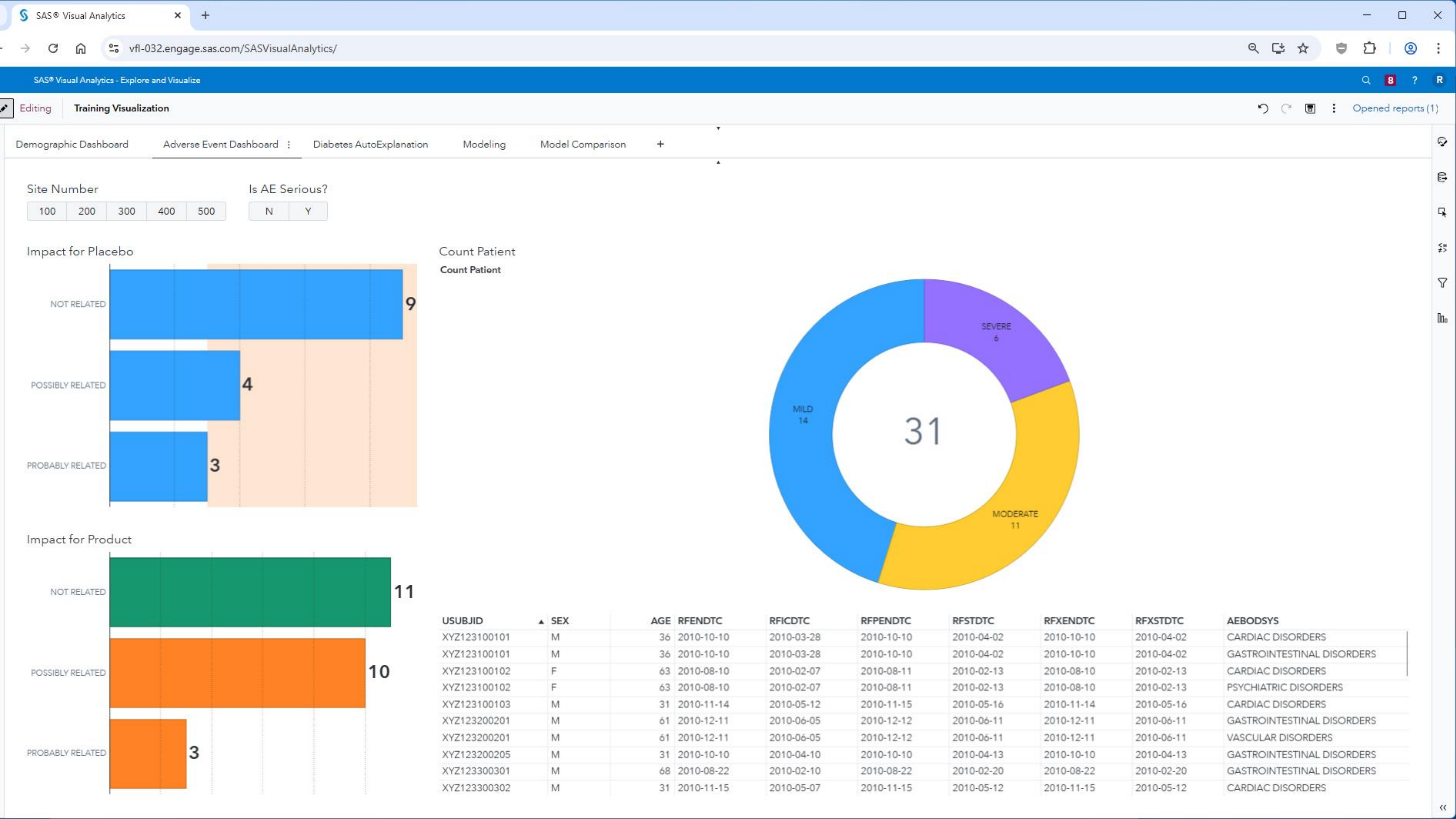
SAS® Visual Analytics - Explore and Visualize
Editing
Training Visualization
Opened reports (1
Demographic Dashboard
Adverse Event Dashboard
Diabetes AutoExplanation
Modeling
Model Comparison
SITEID
Asian
Black or African American
White
Filters: No selections
Frequency
60
ARM
Analgesia HCL 30mg
Placebo
Split by gender
Frequency
40
30
20
10
0
M
F
U
SEX
Age Category Coverage
Frequency
25
20
15
10
5
0
Mid Age
Young
Old People
Age Category
USUBJID SEX AGE RACE ARM ACTARM
XYZ123100101 M 36 Asian Placebo Placebo
XYZ123100102 F 63 White Analgesia HCL 30mg Analgesia HCL 30mg
XYZ123100103 M 31 Black or African American Analgesia HCL 30mg Analgesia HCL 30mg
XYZ123100104 F 37 White Placebo Placebo
XYZ123100105 M 31 Asian Analgesia HCL 30mg Analgesia HCL 30mg
XYZ123100106 F 32 White Placebo Placebo
XYZ123200201 M 61 Asian Analgesia HCL 30mg Analgesia HCL 30mg
XYZ123200202 F 43 White Placebo Placebo

Adverse Event Dashboard



USUBJID	SEX	AGE	RFENDTC	RFICDTC	RFPENDTC	RFSTDTC	RFXENDTC	RFXSTDTC	AEBODSYS
XYZ123100101	M	36	2010-10-10	2010-03-28	2010-10-10	2010-04-02	2010-10-10	2010-04-02	CARDIAC DISORDERS
XYZ123100101	M	36	2010-10-10	2010-03-28	2010-10-10	2010-04-02	2010-10-10	2010-04-02	GASTROINTESTINAL DISORDERS
XYZ123100102	F	63	2010-08-10	2010-02-07	2010-08-11	2010-02-13	2010-08-10	2010-02-13	CARDIAC DISORDERS
XYZ123100102	F	63	2010-08-10	2010-02-07	2010-08-11	2010-02-13	2010-08-10	2010-02-13	PSYCHIATRIC DISORDERS
XYZ123100103	M	31	2010-11-14	2010-05-12	2010-11-15	2010-05-16	2010-11-14	2010-05-16	CARDIAC DISORDERS
XYZ123200201	M	61	2010-12-11	2010-06-05	2010-12-12	2010-06-11	2010-12-11	2010-06-11	GASTROINTESTINAL DISORDERS
XYZ123200201	M	61	2010-12-11	2010-06-05	2010-12-12	2010-06-11	2010-12-11	2010-06-11	VASCULAR DISORDERS
XYZ123200205	M	31	2010-10-10	2010-04-10	2010-10-10	2010-04-13	2010-10-10	2010-04-13	GASTROINTESTINAL DISORDERS
XYZ123300301	M	68	2010-08-22	2010-02-10	2010-08-22	2010-02-20	2010-08-22	2010-02-20	GASTROINTESTINAL DISORDERS
XYZ123300302	M	31	2010-11-15	2010-05-07	2010-11-15	2010-05-12	2010-11-15	2010-05-12	CARDIAC DISORDERS

Effect of the drug in patients with diabetes

Prediction for patients with Diabetes

Automated-Explanation, Logistic Regression, Forest, Decision Tree

SAS® Visual Analytics

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SAS® Visual Analytics - Explore and Visualize

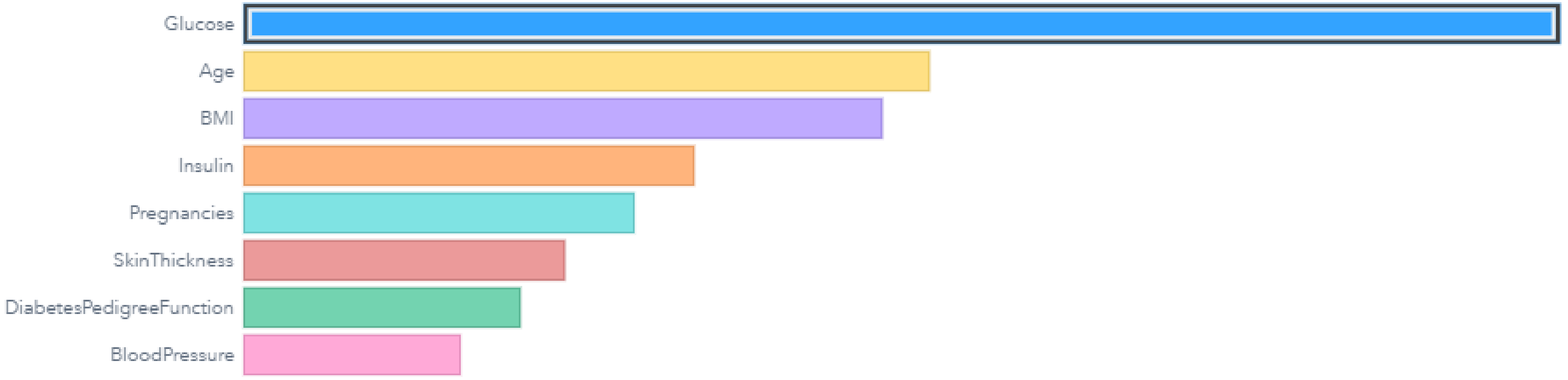
Editing | Training Visualization | Opened reports (1

Demographic Dashboard | Adverse Event Dashboard | Diabetes AutoExplanation | Modeling | Model Comparison

What are the characteristics of Outcome?

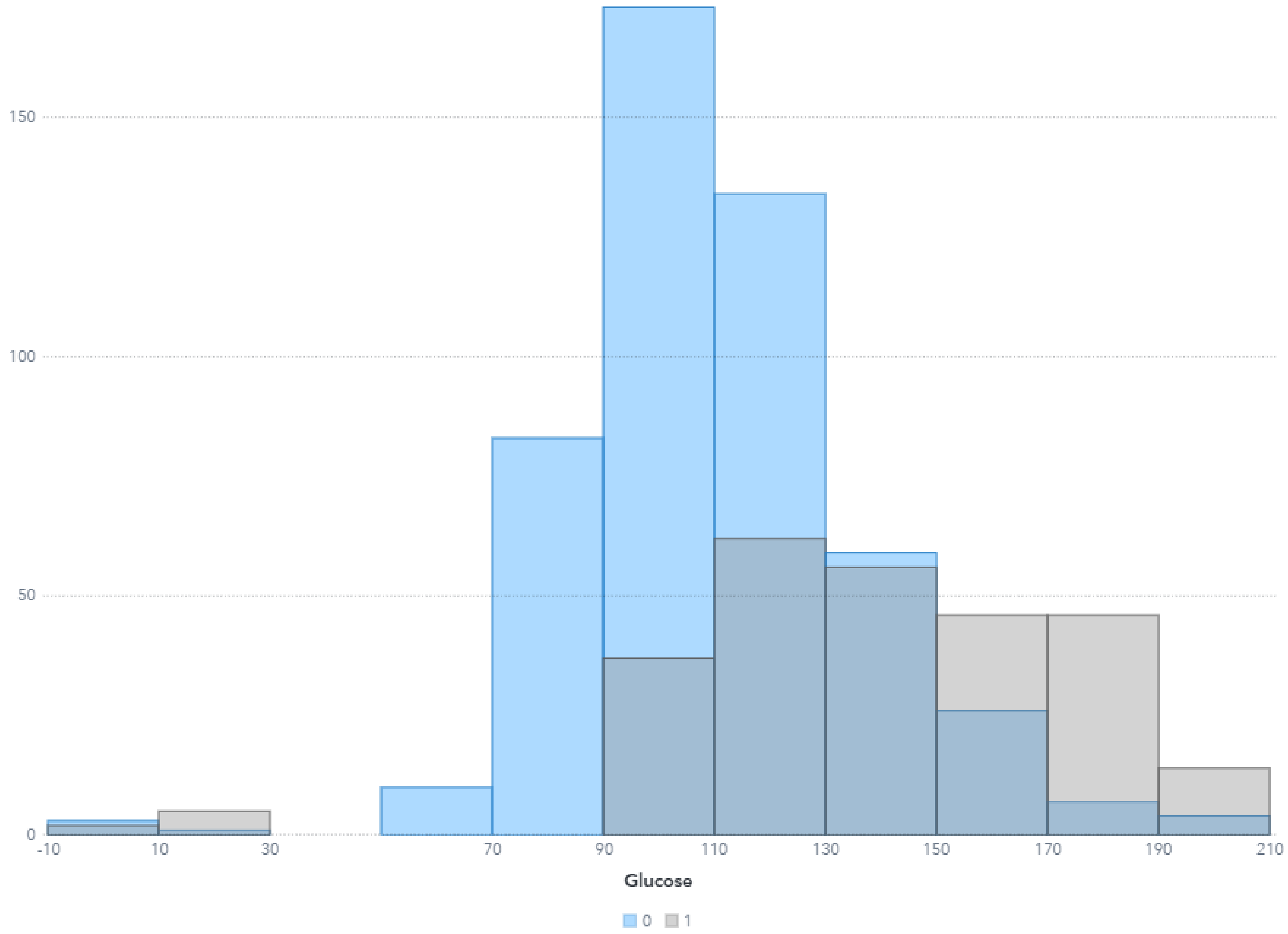
0 is more common at 65.10% (500 of 768). 1 is less common at 34.90%. The three most related factors are Glucose, Age, and BMI.

What factors are most related to Outcome?

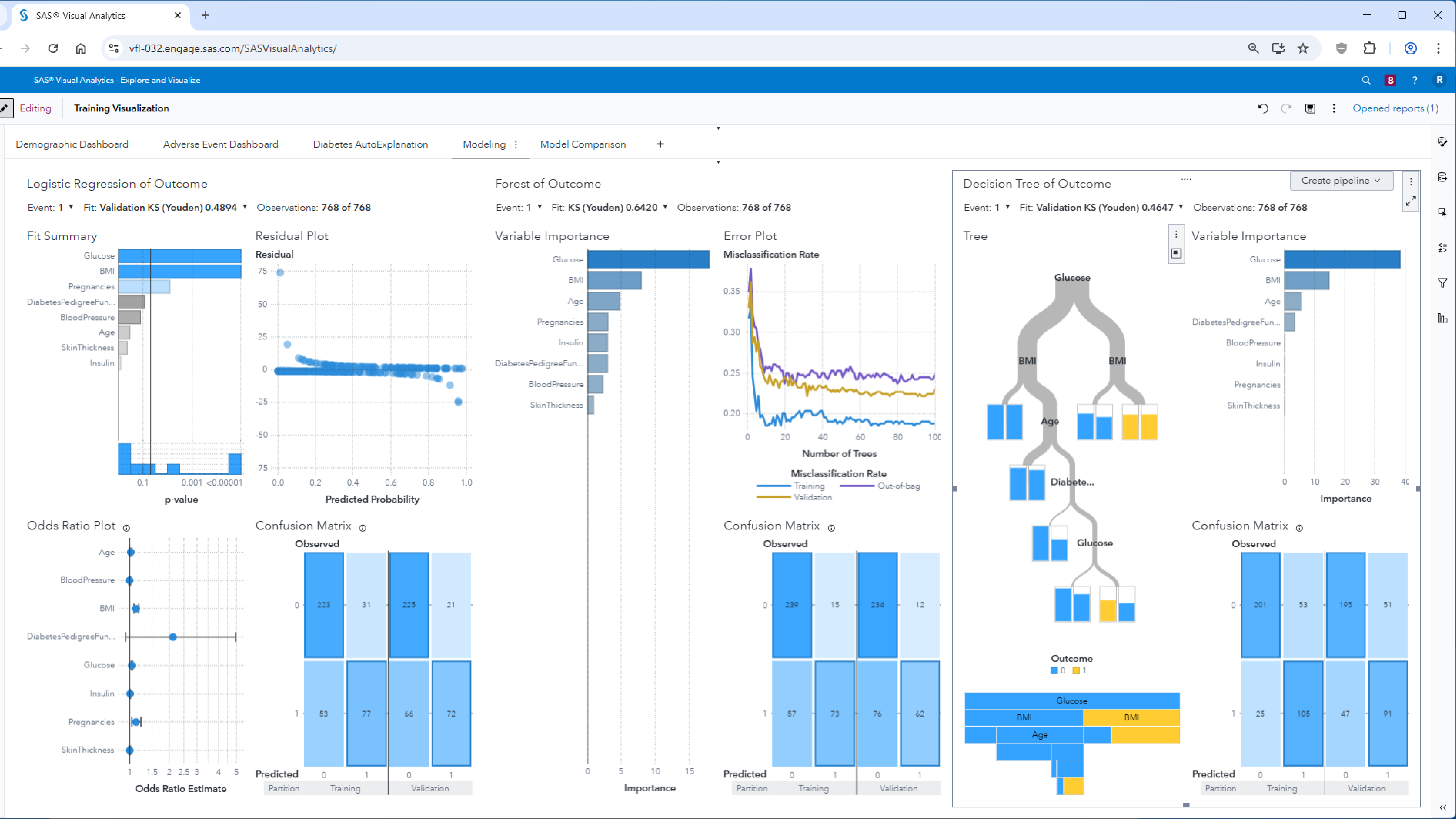


0 | 1

What is the relationship between Outcome and Glucose?



The average Glucose is 110 when Outcome is 0, with a minimum of 0 and a maximum of 197. The average Glucose is 141 when Outcome is 1, with a minimum of 0 and a maximum of 199. Average Glucose is 121, and it ranges from 0 to 199.


SAS® Visual Analytics
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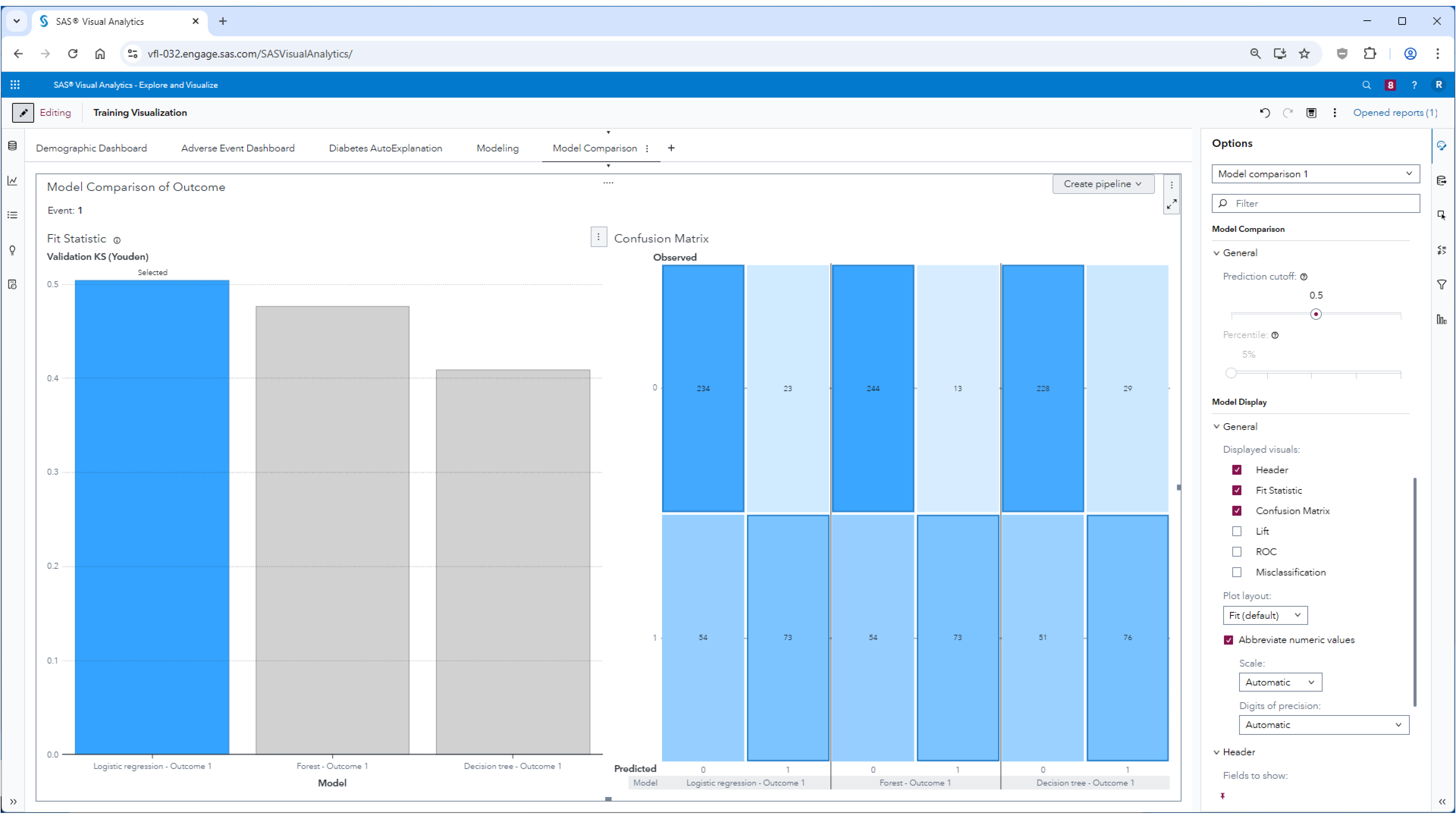
SAS® Visual Analytics - Explore and Visualize
Editing
Training Visualization
Opened reports (1)
Demographic Dashboard
Adverse Event Dashboard
Diabetes AutoExplanation
Modeling
Model Comparison
Logistic Regression of Outcome
Event: 1 Fit: Validation KS (Youden) 0.4894 Observations: 768 of 768
Fit Summary
Glucose
BMI
Pregnancies
DiabetesPedigreeFun...
BloodPressure
Age
SkinThickness
Insulin
0.1
0.001
<0.00001
p-value
Residual Plot
Residual
Predicted Probability
Odds Ratio Plot
Odds Ratio Estimate
Confusion Matrix
Observed
Predicted
Partition
Training
Validation
223
31
225
21
53
77
66
72
Forest of Outcome
Event: 1 Fit: KS (Youden) 0.6420 Observations: 768 of 768
Variable Importance
Importance
Error Plot
Misclassification Rate
Number of Trees
Training
Out-of-bag
Validation
239
15
234
12
57
73
76
62
Decision Tree of Outcome
Create pipeline
Event: 1 Fit: Validation KS (Youden) 0.4647 Observations: 768 of 768
Tree
Glucose
BMI
Age
Diabete...
Outcome
201
53
195
51
25
105
47
91



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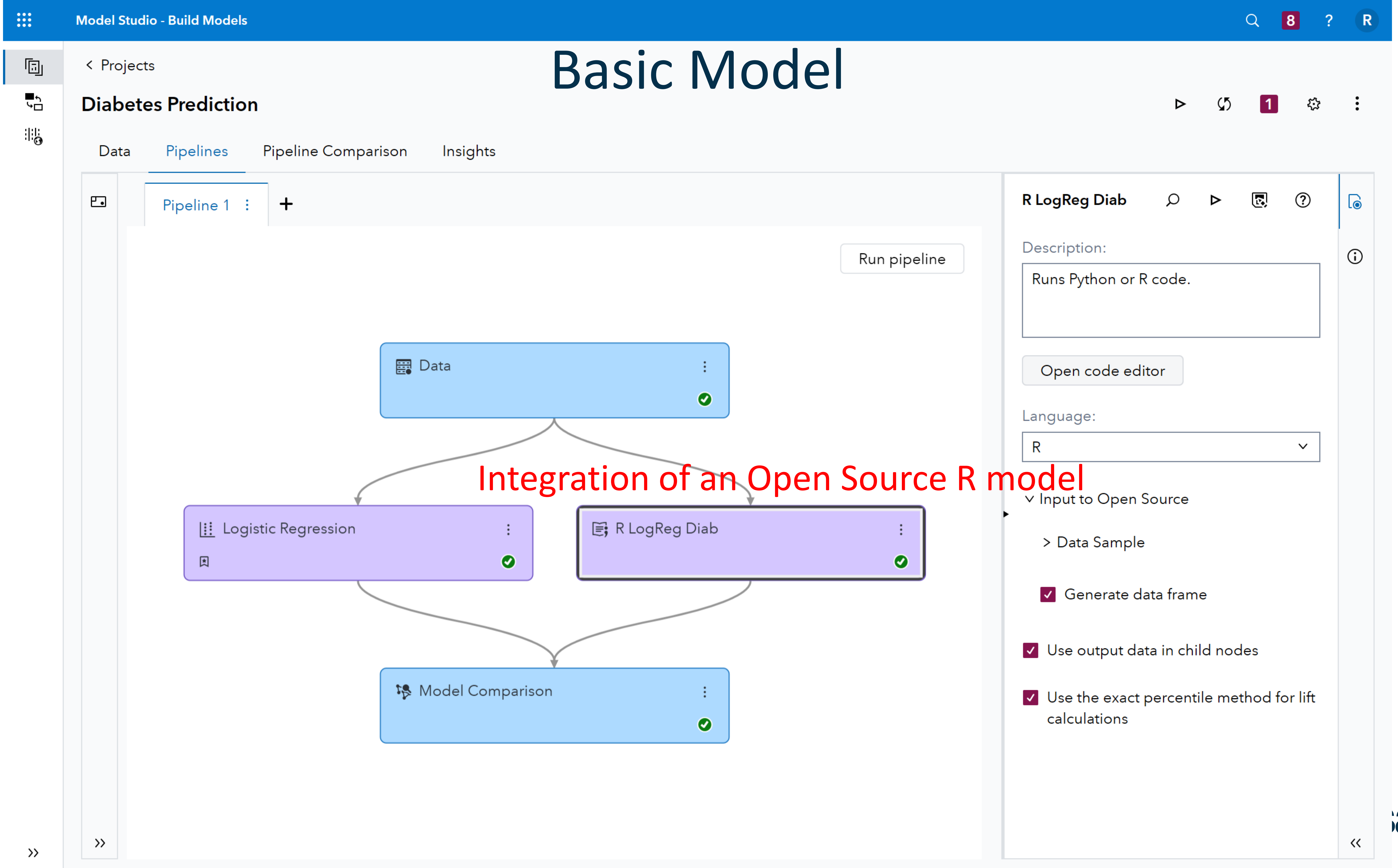
Modeling
Model Comparison
Model Comparison of Outcome
Event: 1
Create pipeline
Fit Statistic
Validation KS (Youden)
Selected
0.5
0.4
0.3
0.2
0.1
0.0
Logistic regression - Outcome 1
Forest - Outcome 1
Decision tree - Outcome 1
Model
Confusion Matrix
Observed
0
1
234
23
244
13
228
29
54
73
54
73
51
76
Predicted
0
1
0
1
0
1
Model
Logistic regression - Outcome 1
Forest - Outcome 1
Decision tree - Outcome 1
Options
Model comparison 1
Filter
Model Comparison
General
Prediction cutoff:
0.5
Percentile:
5%
Model Display
General
Displayed visuals:
Header
Fit Statistic
Confusion Matrix
Lift
ROC
Misclassification
Plot layout:
Fit (default)
Abbreviate numeric values
Scale:
Automatic
Digits of precision:
Automatic
Header
Fields to show:

Advanced Modeling

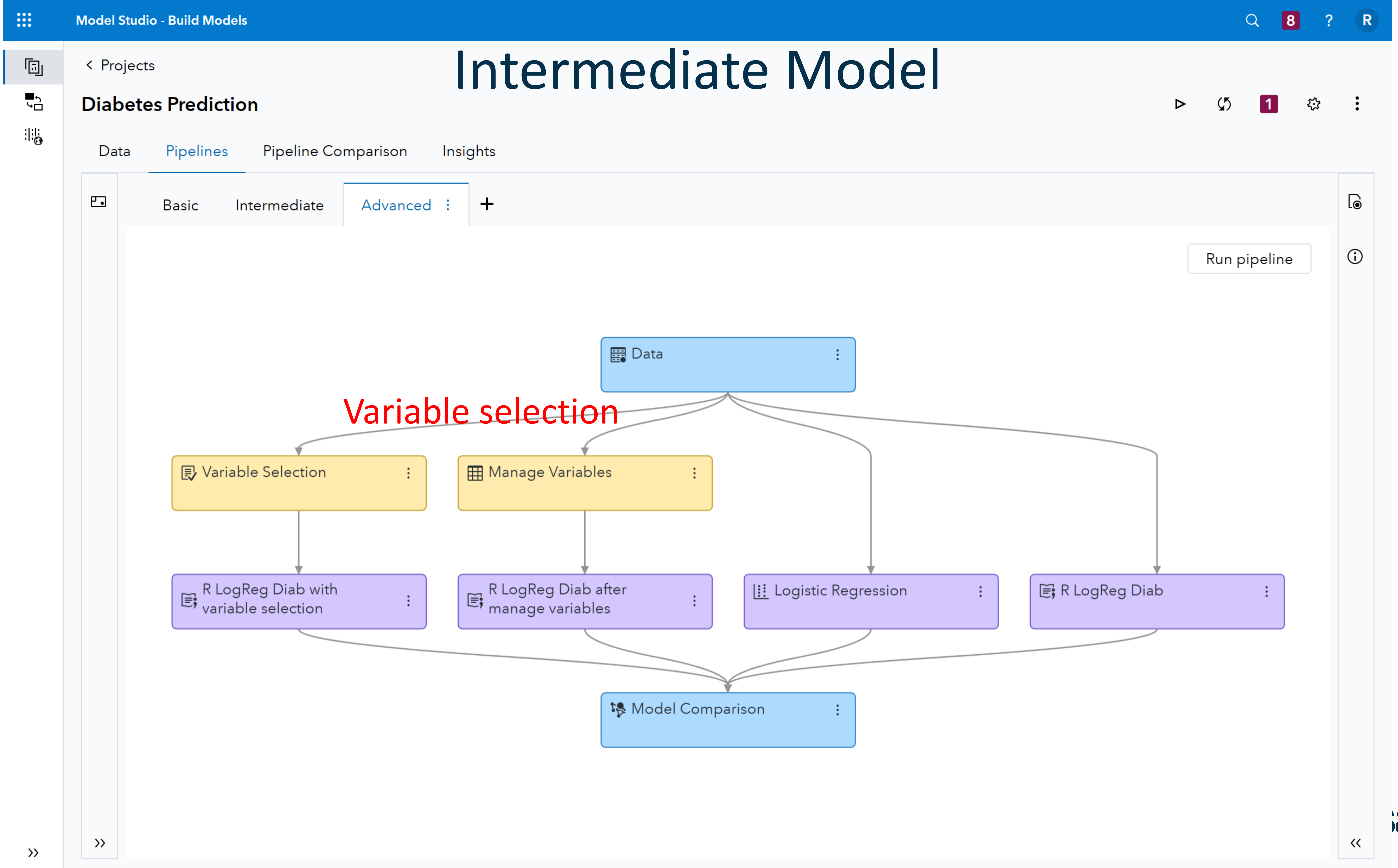
Next future steps, first view - Pipeline Building

Basic Model

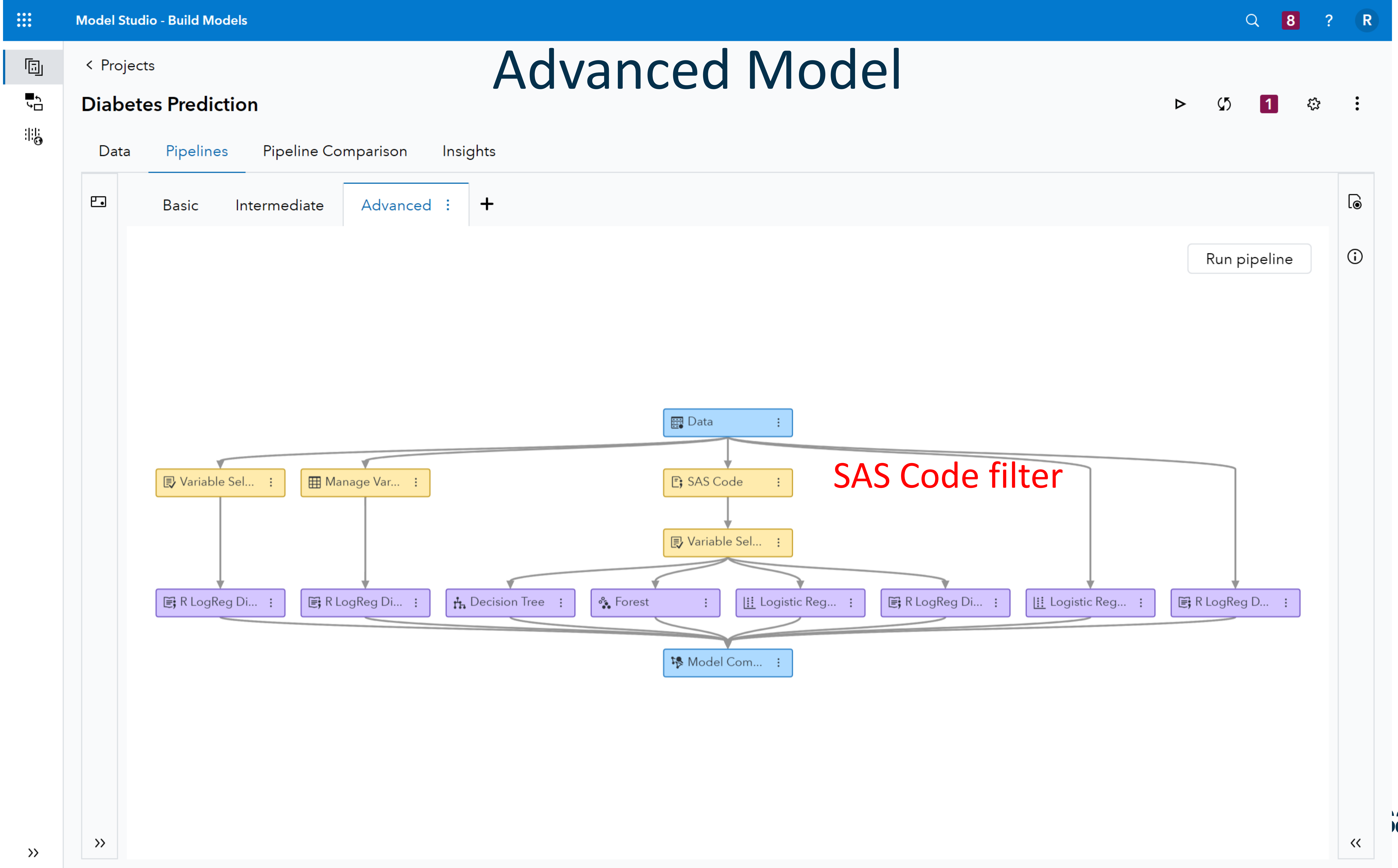


Integration of an Open Source R model

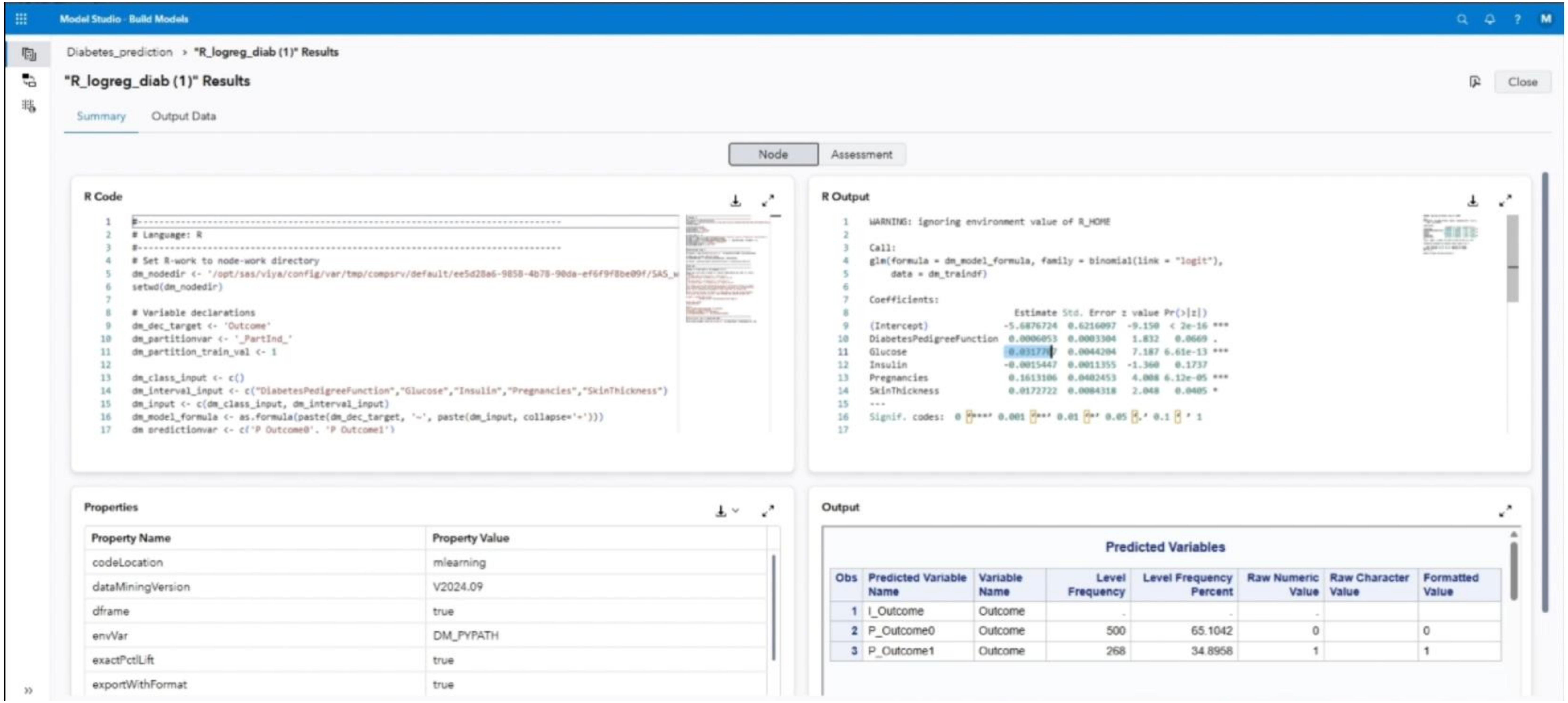
Intermediate Model



Advanced Model

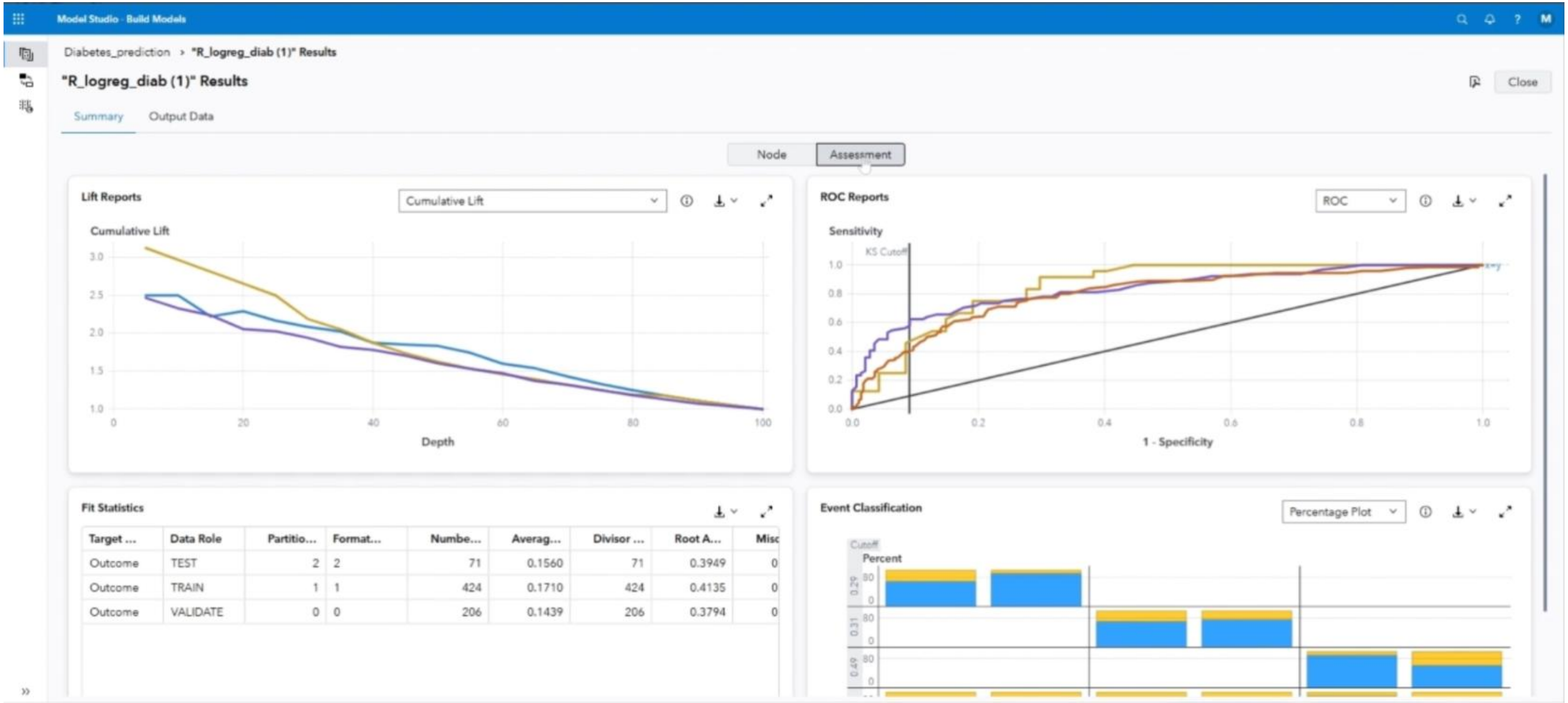


SAS Code filter



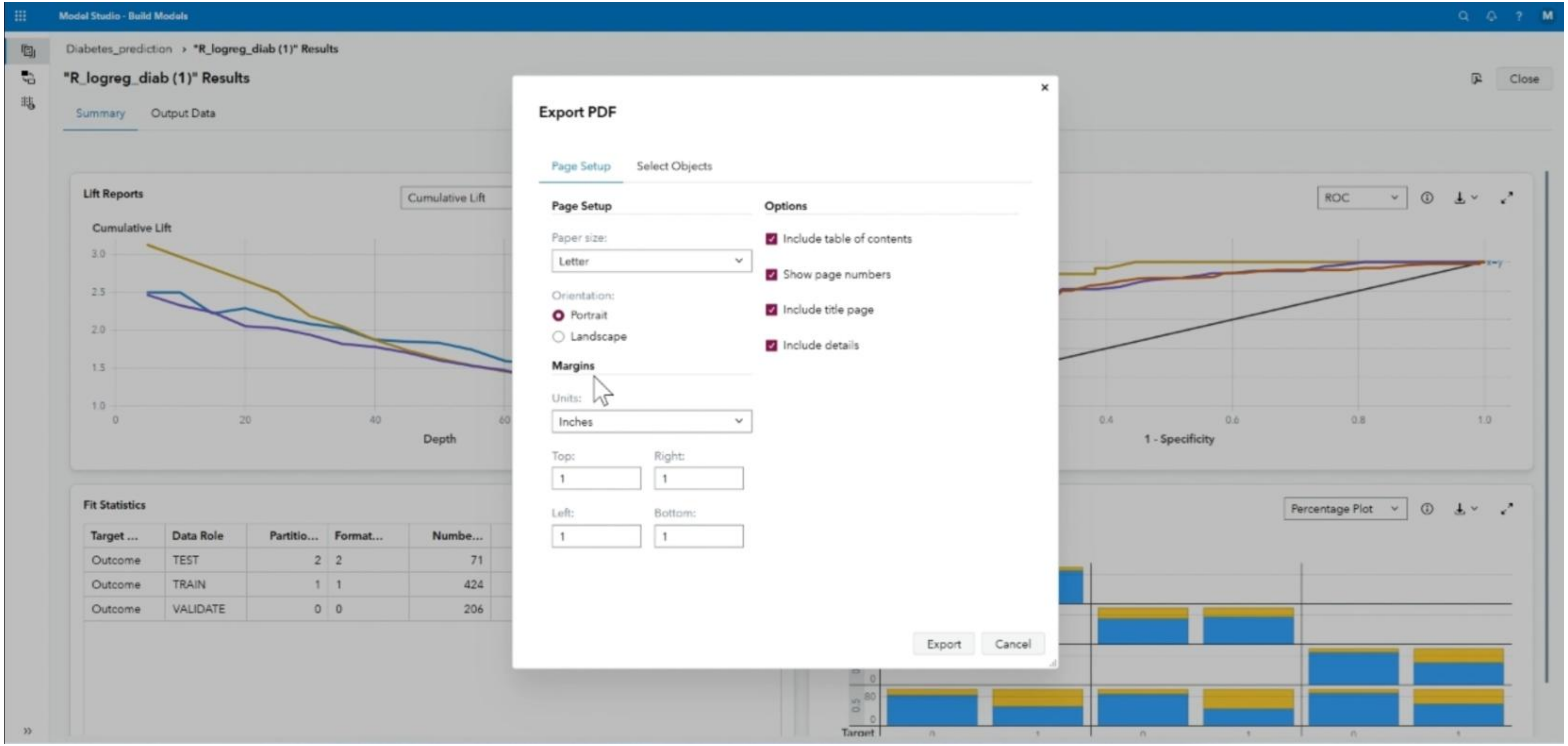
Model Studio · Build Models
Diabetes_prediction > "R_logreg_diab (1)" Results
"R_logreg_diab (1)" Results
Close
Summary
Output Data
Node
Assessment
R Code
#---
Language: R
#---
# Set R-work to node-work directory
dm_nodedir <- '/opt/sas/viya/config/var/tmp/compsrv/default/ee5d28a6-9858-4b78-90da-ef6f9f8be09f/SAS_w
setwd(dm_nodedir)
# Variable declarations
dm_dec_target <- 'Outcome'
dm_partitionvar <- '_PartInd_'
dm_partition_train_val <- 1
dm_class_input <- c()
dm_interval_input <- c("DiabetesPedigreeFunction","Glucose","Insulin","Pregnancies","SkinThickness")
dm_input <- c(dm_class_input, dm_interval_input)
dm_model_formula <- as.formula(paste(dm_dec_target, '~', paste(dm_input, collapse='+')))
dm predictionvar <- c('P Outcome0', 'P Outcome1')
R Output
WARNING: ignoring environment value of R_HOME
Call:
glm(formula = dm_model_formula, family = binomial(link = "logit"),
data = dm_traindf)
Coefficients:
Estimate Std. Error z value Pr(>|z|)
(Intercept) -5.6876724 0.6216097 -9.150 < 2e-16 ***
DiabetesPedigreeFunction 0.0006053 0.0003304 1.832 0.0669 .
Glucose 0.0044204 7.187 6.61e-13 ***
Insulin -0.0015447 0.0011355 -1.360 0.1737
Pregnancies 0.1613106 0.0402453 4.008 6.12e-05 ***
SkinThickness 0.0172722 0.0084318 2.048 0.0405 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Properties
Property Name
Property Value
codeLocation
mlearning
dataMiningVersion
V2024.09
dframe
true
envVar
DM_PYPATH
exactPctlLift
true
exportWithFormat
true
Output
Predicted Variables
Obs
Predicted Variable Name
Variable Name
Level Frequency
Level Frequency Percent
Raw Numeric Value
Raw Character Value
Formatted Value
1 I_Outcome Outcome
2 P_Outcome0 Outcome 500 65.1042 0 0
3 P_Outcome1 Outcome 268 34.8958 1 1



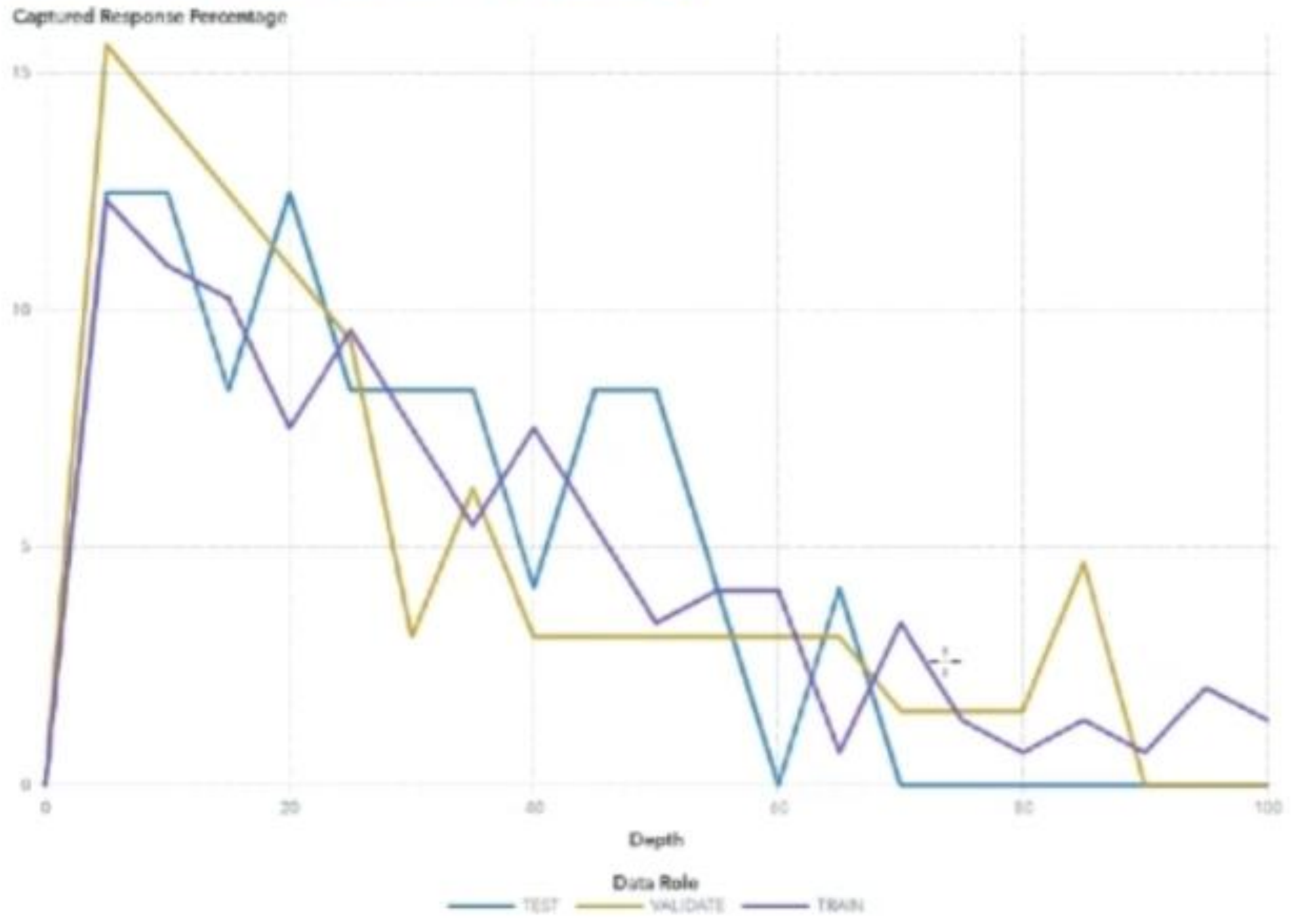
Model Studio - Build Models
Diabetes_prediction › "R_logreg_diab (1)" Results
"R_logreg_diab (1)" Results
Close
Summary
Output Data
Node
Assessment
Lift Reports
Cumulative Lift
Cumulative Lift
Depth
ROC Reports
ROC
Sensitivity
KS Cutoff
1 - Specificity
Fit Statistics
Target ... | Data Role | Partitio... | Format... | Numbe... | Averag... | Divisor ... | Root A... | Misc
Outcome | TEST | 2 | 2 | 71 | 0.1560 | 71 | 0.3949 | 0
Outcome | TRAIN | 1 | 1 | 424 | 0.1710 | 424 | 0.4135 | 0
Outcome | VALIDATE | 0 | 0 | 206 | 0.1439 | 206 | 0.3794 | 0
Event Classification
Percentage Plot
Cutoff
Percent

Model Studio · Build Models
Event Classification
View chart: Percentage Plot
Cutoff
Percent
0.29
0.31
0.49
0.5
80
40
0
Target
Data Role
TEST
TRAIN
VALIDATE
Status CORRECT INCORRECT
The Event Classification report is a visual representation of the confusion matrix at various cutoff values for each partition. The classification cutoffs used in the plot are the default (0.5) and these KS cutoff values for existing partitions: 0.31 (TRAIN), 0.49 (VALIDATE), 0.29 (TEST).
For this data, for the bar corresponding to the event level of Outcome, "1", the segment of the bar colored as "CORRECT" corresponds to true positives.
Close


Model Studio - Build Models
Diabetes_prediction › "R_logreg_diab (1)" Results
"R_logreg_diab (1)" Results
Summary
Output Data
Close
Lift Reports
Cumulative Lift
Depth
ROC
1 - Specificity
Fit Statistics
Target ... | Data Role | Partitio... | Format... | Numbe...
Outcome | TEST | 2 | 2 | 71
Outcome | TRAIN | 1 | 1 | 424
Outcome | VALIDATE | 0 | 0 | 206
Percentage Plot
Export PDF
Page Setup
Select Objects
Page Setup
Paper size:
Letter
Orientation:
Portrait
Landscape
Margins
Units:
Inches
Top:
1
Right:
1
Left:
1
Bottom:
1
Options
Include table of contents
Show page numbers
Include title page
Include details
Export
Cancel

Contents

Captured Response Percentage



At the 5% quantile (depth of 5), the VALIDATE partition has a Captured response percentage of 15.6 (compared to the expected value of 5 for no model). The best possible value of Captured response percentage for this partition at depth 5 is 17.19.

Survival Analysis

SAS® Studio

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SAS® Studio - Develop Code and Flows

New Options View Open Save All

SAS Studio compute conte

Steps

Type to filter list

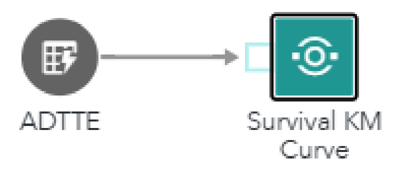
SAS Steps | Shared

- Append Table
- Print Sample (final)
- Survival KM Curve

Start Page | * Flow.flw

Run Cancel Add View

Flow | Generated Code | Submitted Code and Results



Survival KM Curve

Page 1 | Node | Notes

Refresh Edi

Select Time Variable*

AVAL

Select Status Variable*

CNSR

Select Censored Value

0

Select Group Variable*

TRTP

Enter the units of Time

Flow.flw [temp]

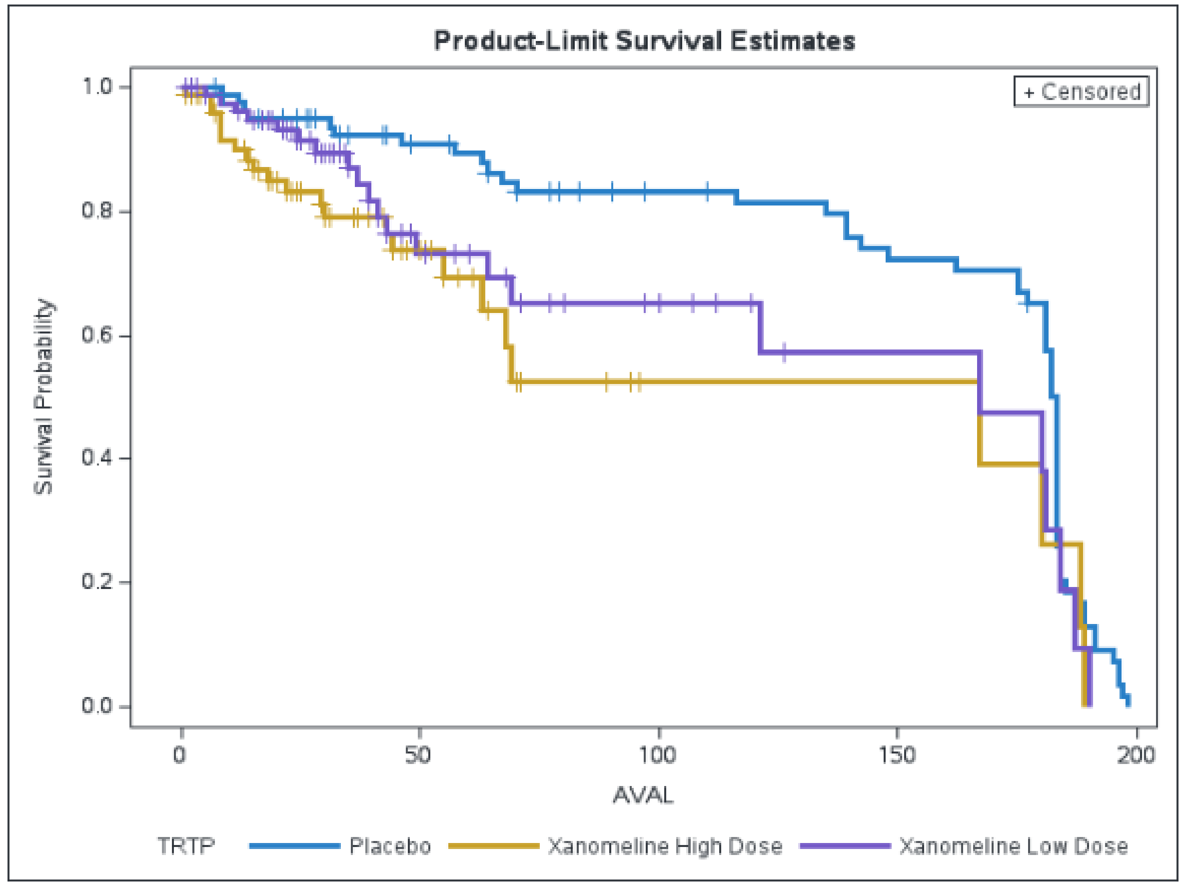
Recover (22) Submission (0)

TRTP	Log-Rank	Wilcoxon
Placebo	-12.151	-1563.0
Xanomeline High Dose	8.452	1396.0
Xanomeline Low Dose	3.699	167.0

Covariance Matrix for the Log-Rank Statistics			
TRTP	Placebo	Xanomeline High Dose	Xanomeline Low Dose
Placebo	17.0228	-7.4077	-9.6152
Xanomeline High Dose	-7.4077	10.9629	-3.5552
Xanomeline Low Dose	-9.6152	-3.5552	13.1704

Covariance Matrix for the Wilcoxon Statistics			
TRTP	Placebo	Xanomeline High Dose	Xanomeline Low Dose
Placebo	307909	-140918	-166991
Xanomeline High Dose	-140918	251039	-110120
Xanomeline Low Dose	-166991	-110120	277111

Test of Equality over Strata			
Test	Chi-Square	DF	Pr > Chi-Square
Log-Rank	9.9667	2	0.0069
Wilcoxon	10.4177	2	0.0055
-2Log(LR)	1.2968	2	0.5229



Appendix / References

For more information on

- ADTTE, please refer to the document "Adam_tte_final_v1.pdf"
- Data Dictionary: Complete data description
- Custom Steps, refer to this Video
- Building Beautiful Reports: Guidance on how to create a compelling data story by designing a report that is both visually appealing and easy to understand.
- SAS Solution: SAS Clinical Acceleration

Download the Use Case Demo Data

Use Case sources on Github or here in the Chat

- ADTTE.csv
- AE_DM.csv
- DIABETES.csv
- DM_DM.csv
- Diabetes Prediction (ORG).zip
- R Code.txt
- README.md
- SAS Code.txt
- Survival KM Curve.step
- registration-guide-skill-builder-for-students.pdf

Link Github

https://github.com/BRERAS/SAS-LS-Analytics-School/archive/refs/heads/main.zip

Let's start the Practice

Log in to SAS

https://learn.sas.com